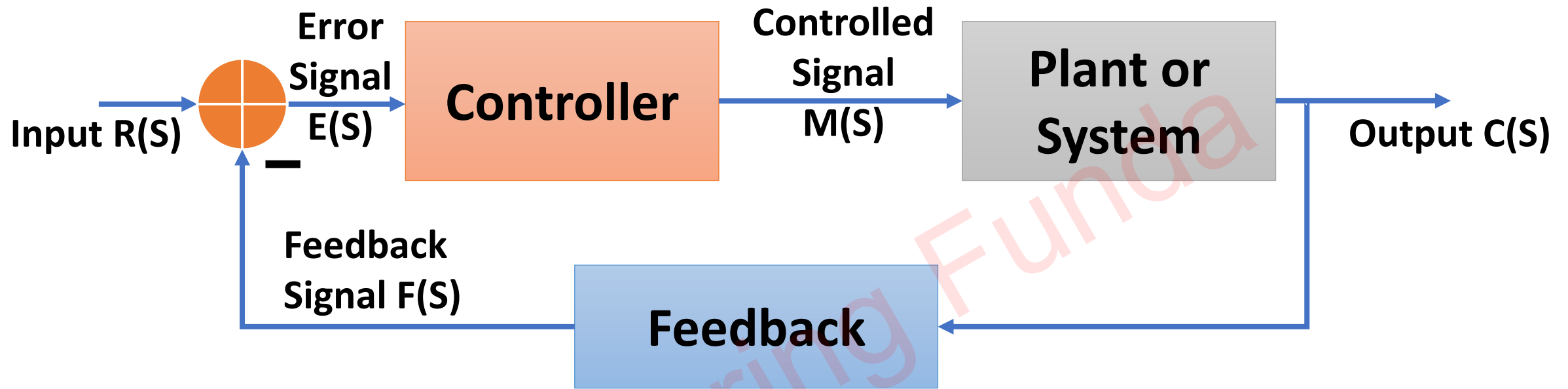


Proportional Controller

Outlines

- ❑ Basics of Proportional Controller
- ❑ Output of Proportional Controller
- ❑ Physical Understanding of Proportional Controller
- ❑ Significance of Proportional Controller
- ❑ Block Diagram of Proportional Controller
- ❑ Transfer Function of Proportional Controller
- ❑ Pros and Cons of Proportional Controller

Basics of Proportional Controller



❑ If the Controller is a Proportional Controller, then the Gain of controller is constant K_P .

❑ Let, the System Gain is $G(S)$ and Feedback is $H(S)$, So Output of Proportional Controller is given by:

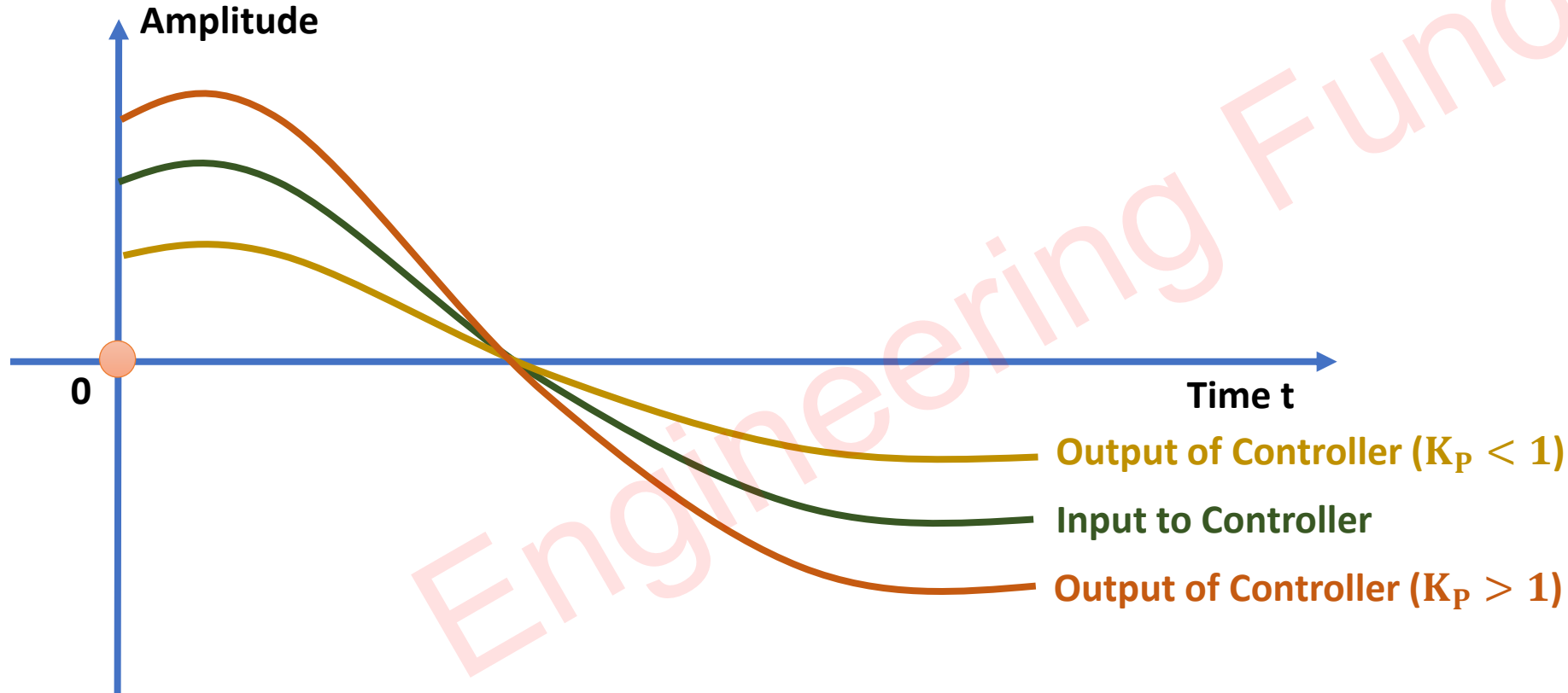
$$\Rightarrow M(S) = K_P E(S) = K_P (R(S) - F(S)) = K_P (R(S) - C(S)H(S))$$

❑ Transfer Function $T(S)$ of the system is given by:

$$\Rightarrow T(S) = \frac{C(S)}{R(S)} = \frac{K_P G(S)}{1 + K_P G(S)H(S)}$$

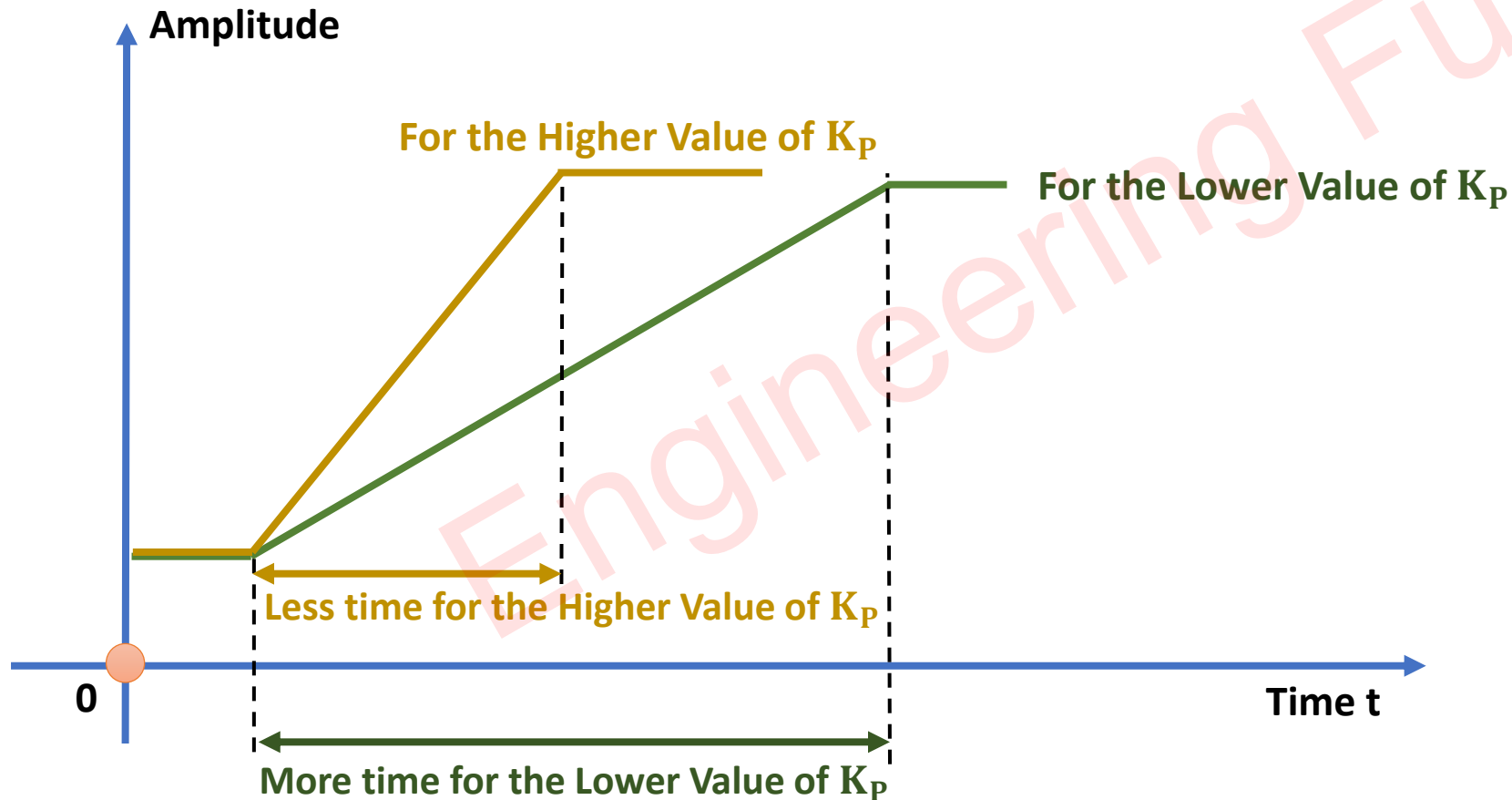
Physical Understanding of Proportional Controller

- ❑ The Gain of Proportional controller is constant K_P .
- ❑ The shape of the Output of the Proportional Controller is the same as the input to the controller, Only the Output is multiplied by the constant K_P .

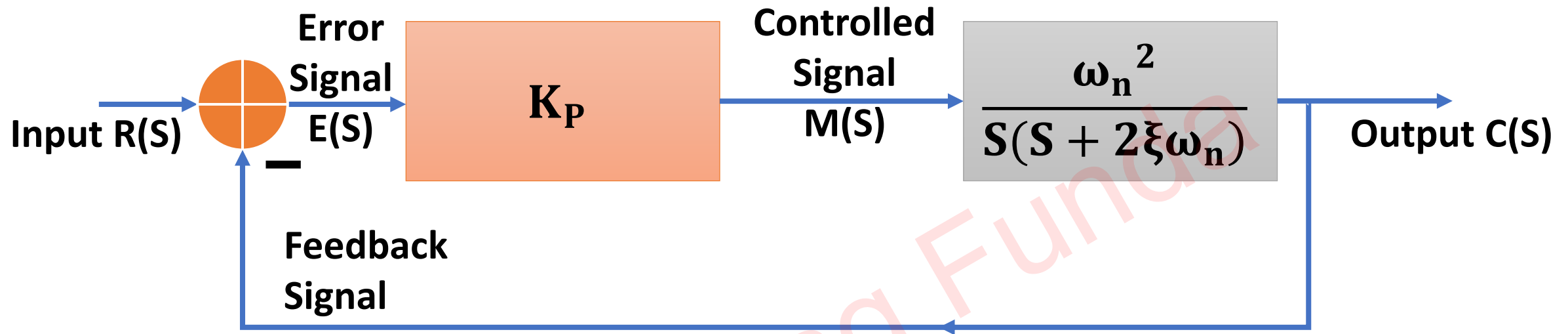


Significance of Proportional Controller

- ❑ For higher values of K_P , the Response is faster, which means it takes less time to get change in output.
- ❑ For lower values of K_P , the Response is slower, which means it takes more time to get change in output.



Block Diagram of Proportional Controller for standard 2nd Order System



□ Transfer Function $T(S)$ of the system:

$$\Rightarrow T(S) = \frac{C(S)}{R(S)} = \frac{G(S)}{1 + G(S)H(S)}$$

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{K_P \omega_n^2}{S^2 + 2\xi\omega_n S + K_P \omega_n^2}$$

- K_P affects the natural Resonance.
- Higher the value of K_P , Higher the Natural Resonance.
- Lower the value of K_P , Lower the Natural Resonance.

□ Transfer Function of standard 2nd Order System:

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{\omega_n^2}{S^2 + 2\xi\omega_n S + \omega_n^2}$$

Advantages of Proportional Controller

- ☐ Simplicity {No Integral/Derivative}.
- ☐ Stability in Simple System.
- ☐ For higher value of K_P , it gives faster response.

Disadvantages of Proportional Controller

- ☐ Steady State Error.
- ☐ Overshoot at higher Gain.
- ☐ Slow Response to Small Errors.
- ☐ Sensitive to change in load.
- ☐ Can not handle nonlinearities.

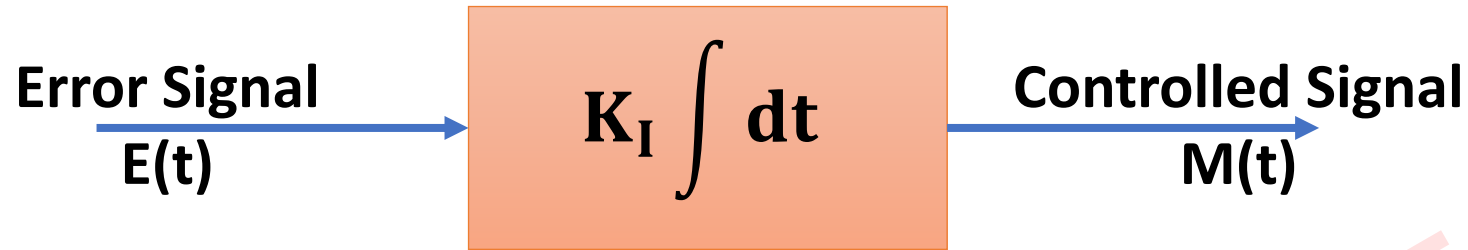
Integral Controller

Outlines

- ☐ Basics of Integral Controller
- ☐ Output of Integral Controller
- ☐ Block Diagram of Integral Controller
- ☐ Transfer Function of Integral Controller
- ☐ Physical Understanding of Integral Controller
- ☐ Significance of Integral Controller
- ☐ Pros and Cons of Integral Controller

Basics of Integral Controller

❑ Integral Controller produces an Output which is an integration of Input Signal.



❑ Output of Integral Controller $M(t)$ is given by:

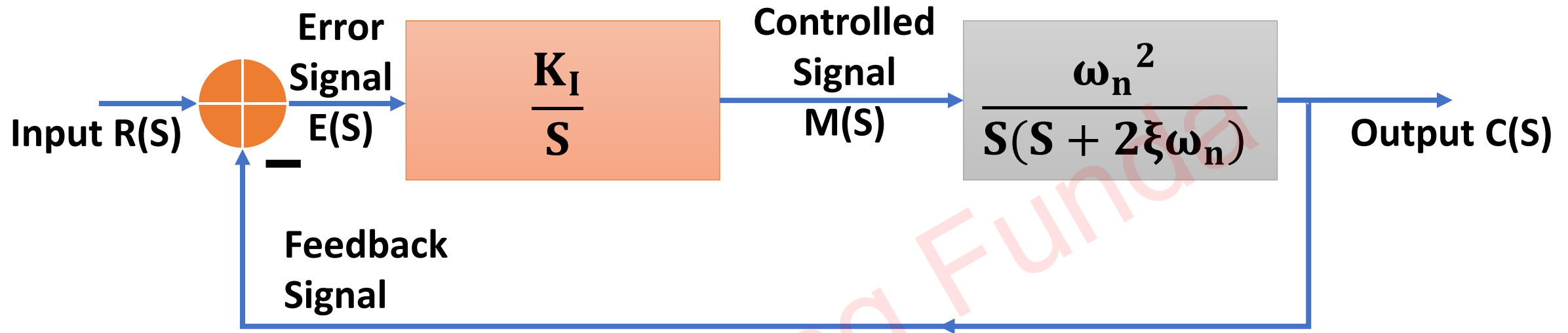
$$\Rightarrow M(t) = K_I \int E(t) dt$$

❑ By Taking Laplace Transform:

$$\Rightarrow M(S) = K_I E(S) / S$$

$$\Rightarrow \frac{M(S)}{E(S)} = \frac{K_I}{S}$$

Block Diagram of Integral Controller for standard 2nd Order System



□ Transfer Function $T(S)$ of the system:

$$\Rightarrow T(S) = \frac{C(S)}{R(S)} = \frac{G(S)}{1 + G(S)H(S)}$$

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{K_I \omega_n^2}{S^3 + 2\xi\omega_n S^2 + K_I \omega_n^2}$$

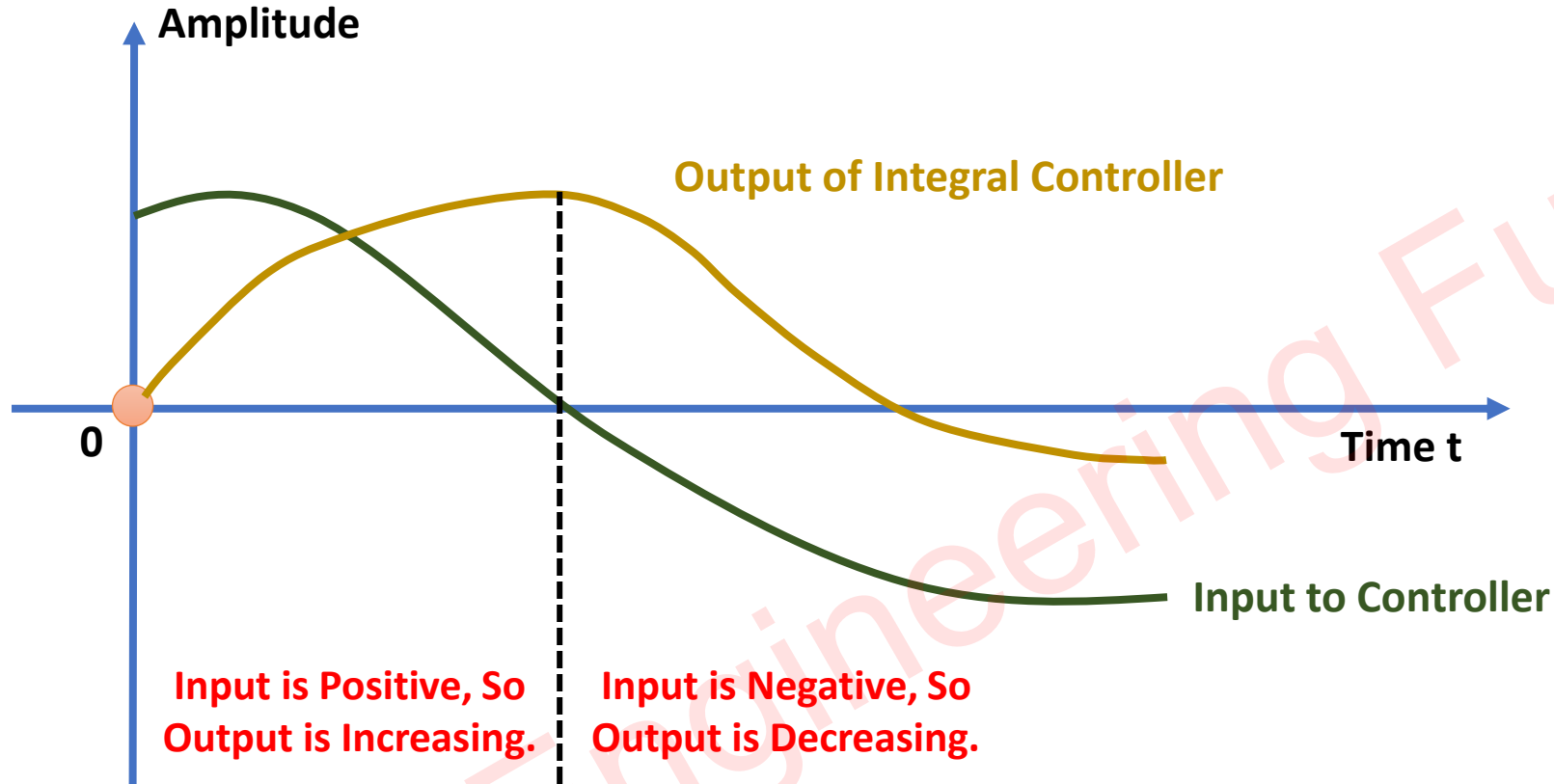
- Integral Controller adds one pole to the system. Which reduces the stability of the system.
- But Integral Controller eliminates steady-state error.

□ Transfer Function of standard 2nd Order System:

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{\omega_n^2}{S^2 + 2\xi\omega_n S + \omega_n^2}$$

Physical Understanding of Integral Controller

□ Integration means summing up input signal with respect to time.



□ As Integration Output is based on summing up the Input signal, It eliminates the steady-state error.

Advantages of Integral Controller

- ☐ Elimination of Steady State Error.
- ☐ Improved System Accuracy.
- ☐ Compensation for Constant Disturbances.

Disadvantages of Integral Controller

- ☐ Slower Response.
- ☐ Reduced Stability.
- ☐ Takes time to get stabilized Output.

Engineering Funda

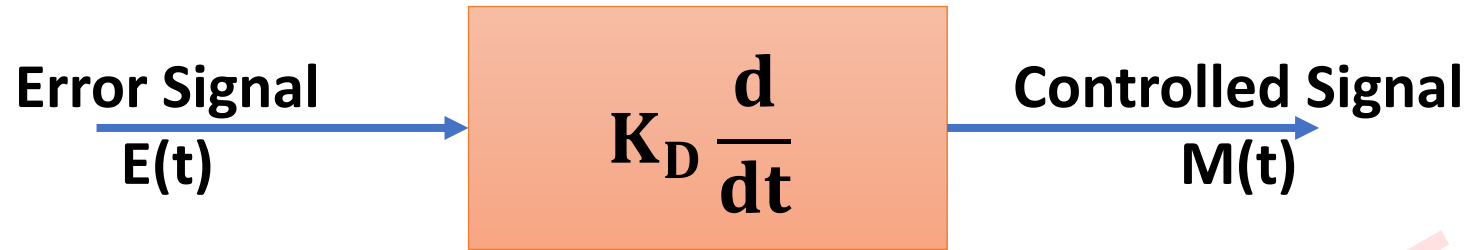
Derivative Controller

Outlines

- ☐ Basics of Derivative Controller
- ☐ Output of Derivative Controller
- ☐ Block Diagram of Derivative Controller
- ☐ Transfer Function of Derivative Controller
- ☐ Physical Understanding of Derivative Controller
- ☐ Significance of Derivative Controller
- ☐ Pros and Cons of Derivative Controller

Basics of Derivative Controller

❑ Derivative Controller produces an Output which is a Differentiation of Input Signal.



❑ Output of Integral Controller $M(t)$ is given by:

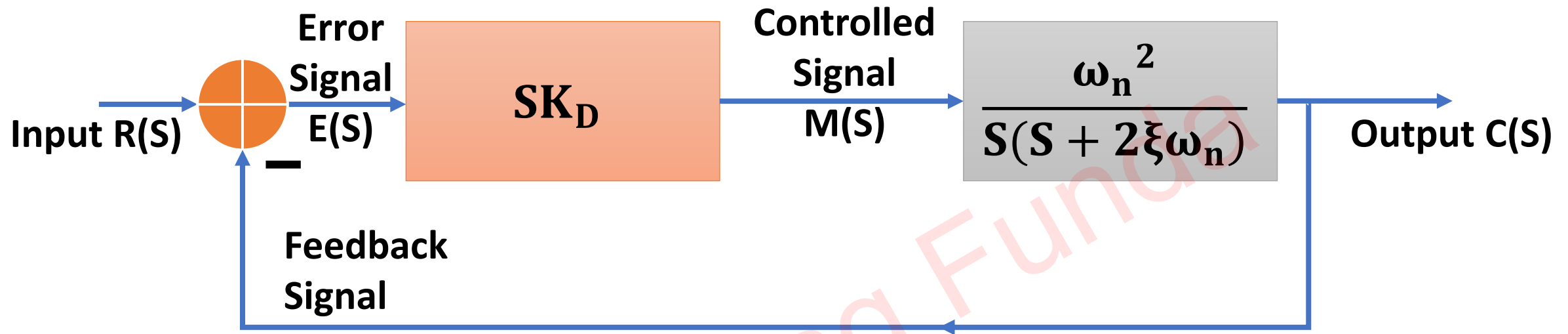
$$\Rightarrow M(t) = K_D \frac{dE(t)}{dt}$$

❑ By Taking Laplace Transform:

$$\Rightarrow M(S) = SK_D E(S)$$

$$\Rightarrow \frac{M(S)}{E(S)} = SK_D$$

Block Diagram of Derivative Controller for standard 2nd Order System



□ Transfer Function $T(S)$ of the system:

$$\Rightarrow T(S) = \frac{C(S)}{R(S)} = \frac{G(S)}{1 + G(S)H(S)}$$

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{K_D \omega_n^2}{S + 2\xi\omega_n + K_D \omega_n^2}$$

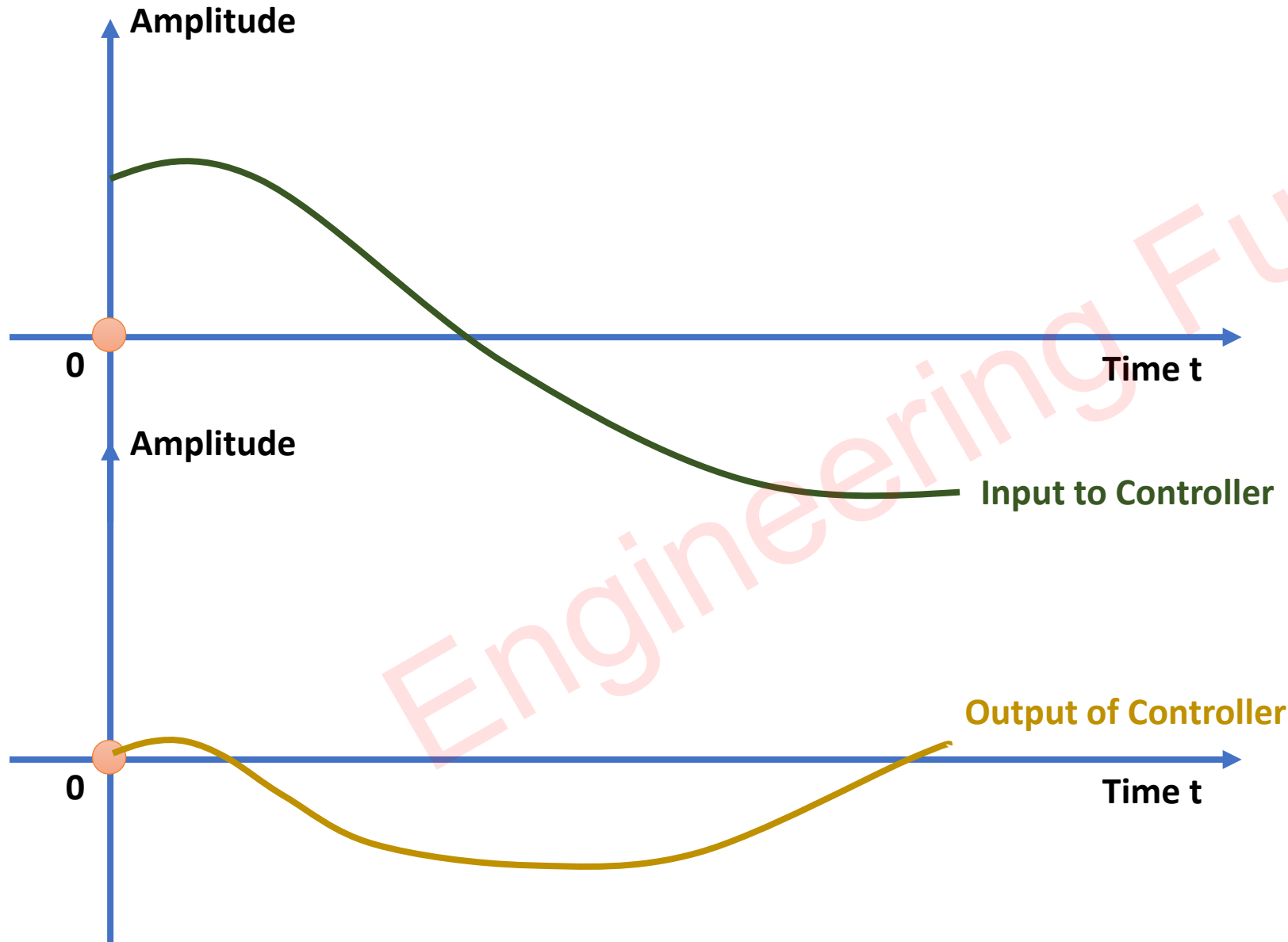
- Derivative Controller adds one zero to the system. Which improves the stability of the system.
- But the Derivative Controller can not eliminate steady-state error.

□ Transfer Function of standard 2nd Order System:

$$\Rightarrow \frac{C(S)}{R(S)} = \frac{\omega_n^2}{S^2 + 2\xi\omega_n S + \omega_n^2}$$

Physical Understanding of Derivative Controller

❑ Derivative means Rate of change of input signal with respect to time {Slope of Input}.



Advantages of Derivative Controller

- ☐ Improved Stability
- ☐ Faster Response
- ☐ Reduced Overshoot
- ☐ Less time to get stabilized Output

Disadvantages of Derivative Controller

- ☐ Sensitive to High Frequency Noise
- ☐ Steady State Error
- ☐ Limited Useful in Slower Systems
- ☐ Tuning is Challenging

Engineering Funda

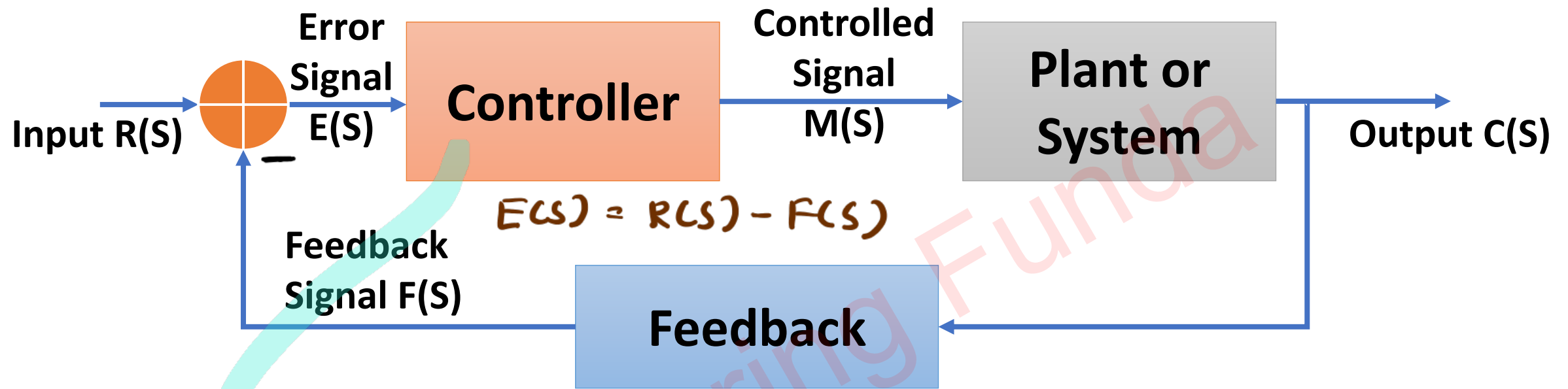
PI Controller

Outlines

- ❑ Basics of PI Controller
- ❑ Output of PI Controller
- ❑ Block Diagram of PI Controller
- ❑ Transfer Function of PI Controller
- ❑ Physical Understanding of PI Controller
- ❑ Significance of PI Controller
- ❑ Pros and Cons of PI Controller

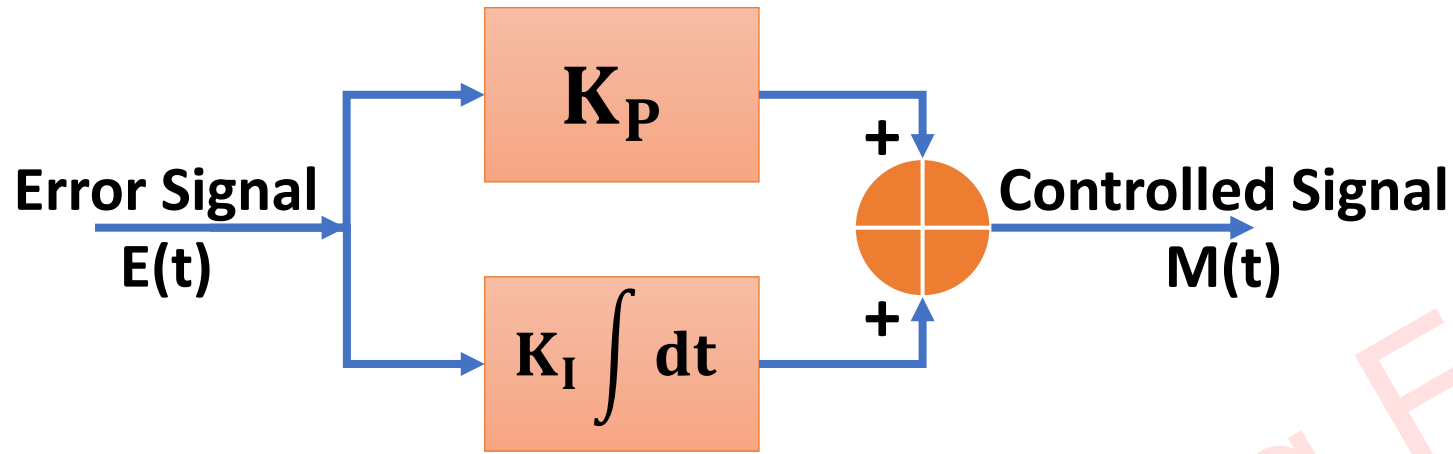
Engineering Funda

Basics of PI Controller



PI Controller
Proportional Controller
Integral Controller.

Output of PI Controller



~> Output of PI Controller is given by:

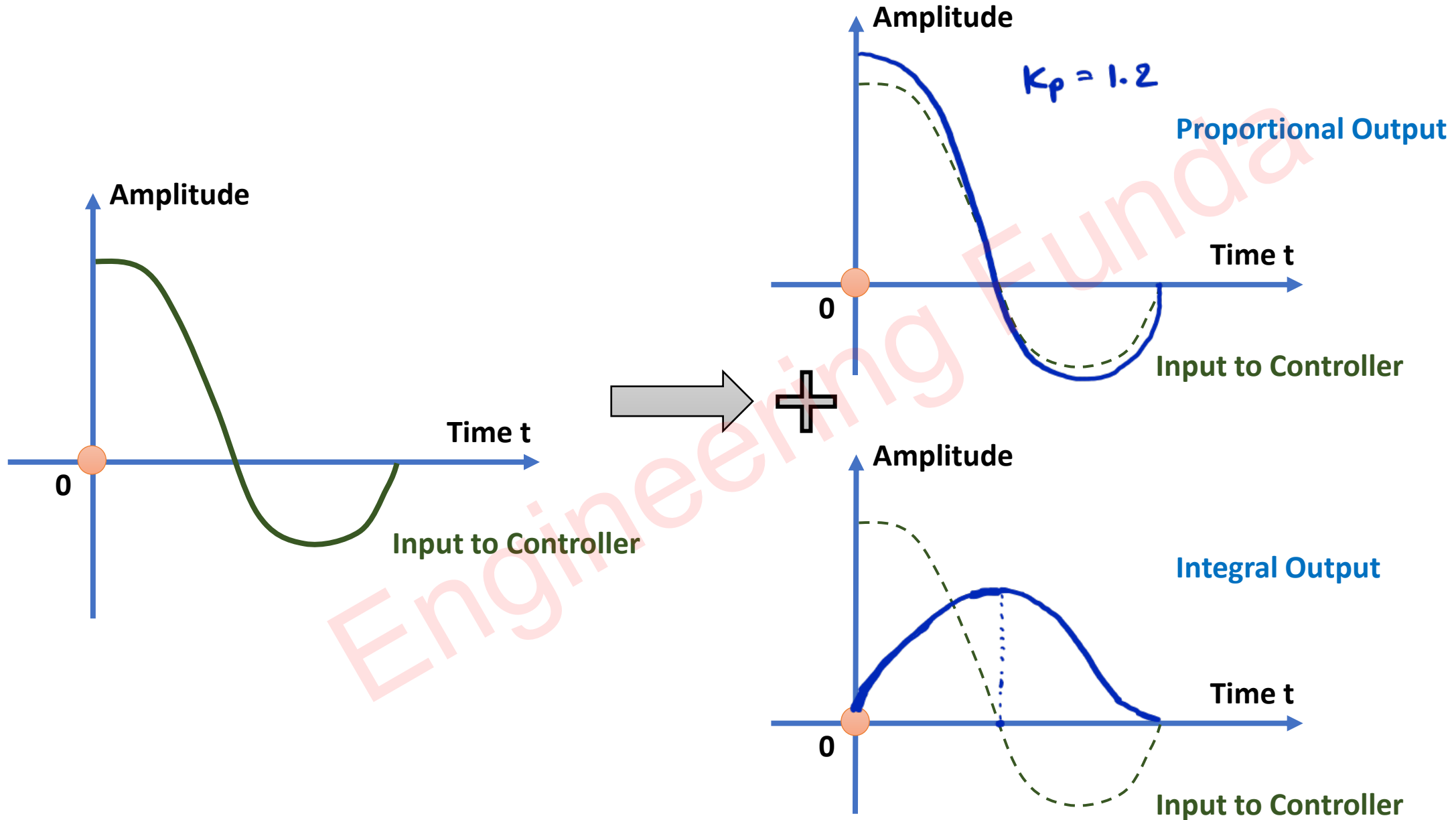
$$\Rightarrow m(t) = K_P E(t) + K_I \int E(t) dt$$

~> Apply Laplace X'form.

$$\Rightarrow M(s) = K_P E(s) + K_I E(s)/s$$

$$\Rightarrow \frac{M(s)}{E(s)} = K_P + \frac{K_I}{s}$$

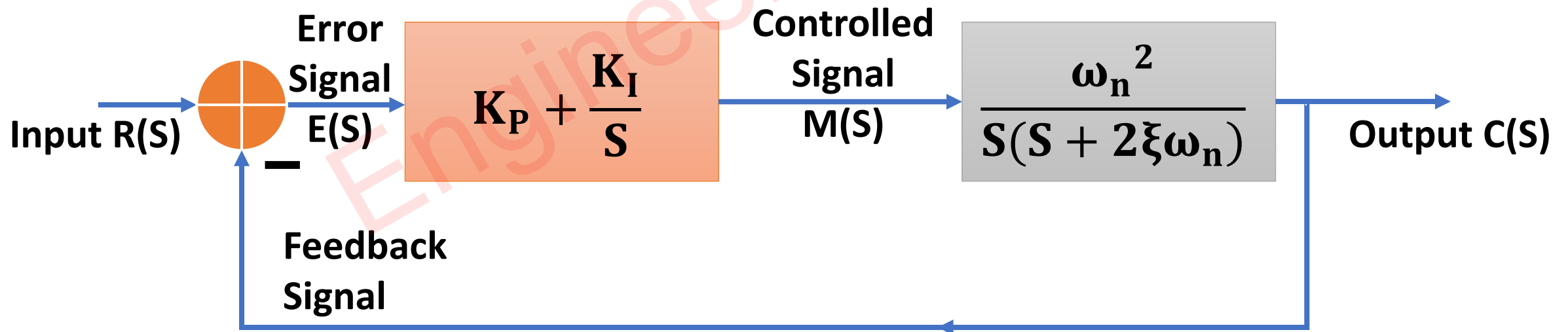
Physical Understanding of PI Controller



Significance of PI Controller

- ❑ PI Controller Eliminates Steady State Error {Due to Integral Controller}.
- ❑ PI Controller offers Better Stability compared to Only Integral Controller.
- ❑ PI Controller has improved Damping {Compared to P and PD Controller}.
- ❑ PI Controller has a lower Peak Overshoot.
- ❑ Tuning of PI Controller is Easy {Ziegler-Nichols Tuning or Manual Tuning}.
- ❑ The Implementation of the PI Controller is easier than that of the PD and PID Controllers.

Block Diagram of PI Controller with 2nd Order System



⇒ Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{K(s)}{1 + K(s)H(s)}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{\frac{(K_1 + sK_p)\omega_n^2}{s^2(s + 2\xi\omega_n)}}{1 + \frac{(K_1 + sK_p)\omega_n^2}{s^2(s + 2\xi\omega_n)}}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{(K_1 + K_p s)\omega_n^2}{s^2(s + 2\xi\omega_n) + (K_1 + sK_p)\omega_n^2}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{(K_1 + K_p s)\omega_n^2}{s^3 + 2\xi\omega_n s^2 + K_p\omega_n^2 s + K_1\omega_n^2}$$

$$\Rightarrow H(s) = 1$$

$$G(s) = \left(K_p + \frac{K_I}{s} \right) \left(\frac{\omega_n^2}{s(s + 2\xi\omega_n)} \right)$$

$$\Rightarrow K(s) = \frac{(K_1 + sK_p)\omega_n^2}{s^2(s + 2\xi\omega_n)}$$

Advantages of PI Controller

- ☐ Improved Damping
- ☐ Lower Overshoot
- ☐ Zero OFFSET {Zero Steady State Error}
- ☐ Improved Accuracy

Disadvantages of PI Controller

- ☐ Increase in Rise Time
- ☐ Slower Response
- ☐ Not Effective in Oscillatory Systems

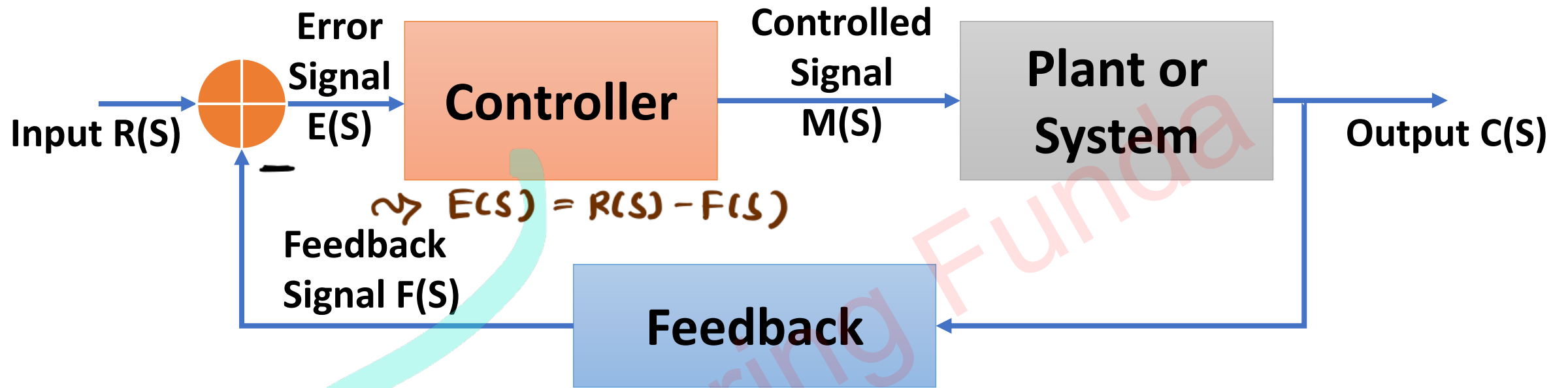
Engineering Funda

PD Controller

Outlines

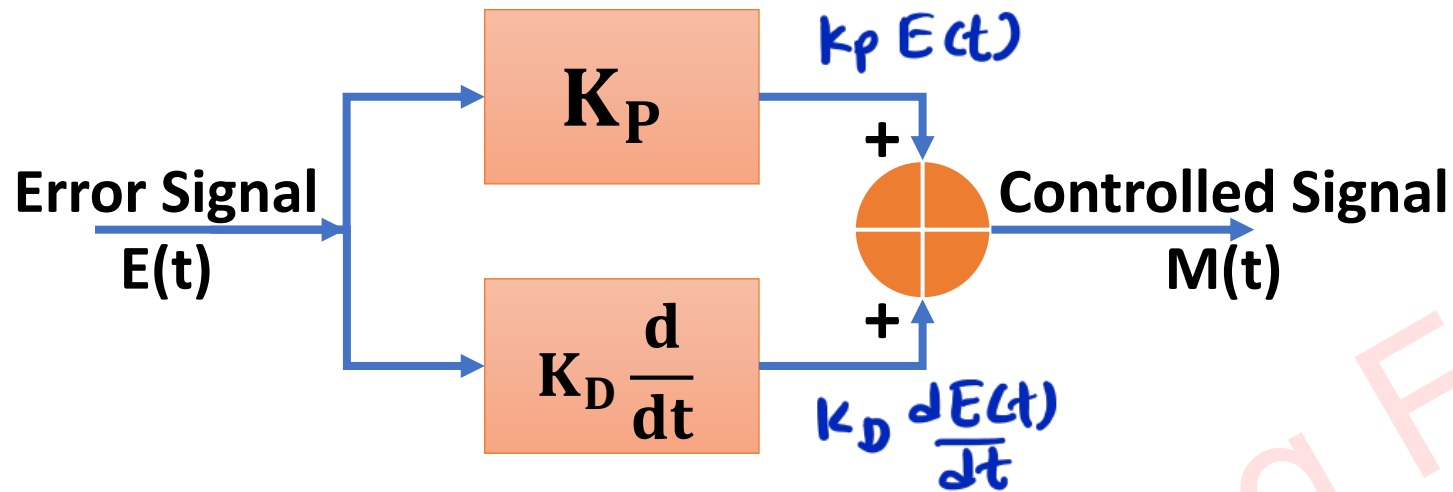
- ☐ Basics of PD Controller
- ☐ Output of PD Controller
- ☐ Block Diagram of PD Controller
- ☐ Transfer Function of PD Controller
- ☐ Physical Understanding of PD Controller
- ☐ Significance of PD Controller
- ☐ Pros and Cons of PD Controller

Basics of PD Controller



PD Controller
Proportional Controller
Derivative Controller.

Output of PD Controller



⇒ Output of PD Controller is given by :

$$\Rightarrow m(t) = K_P E(t) + K_D \frac{dE(t)}{dt}$$

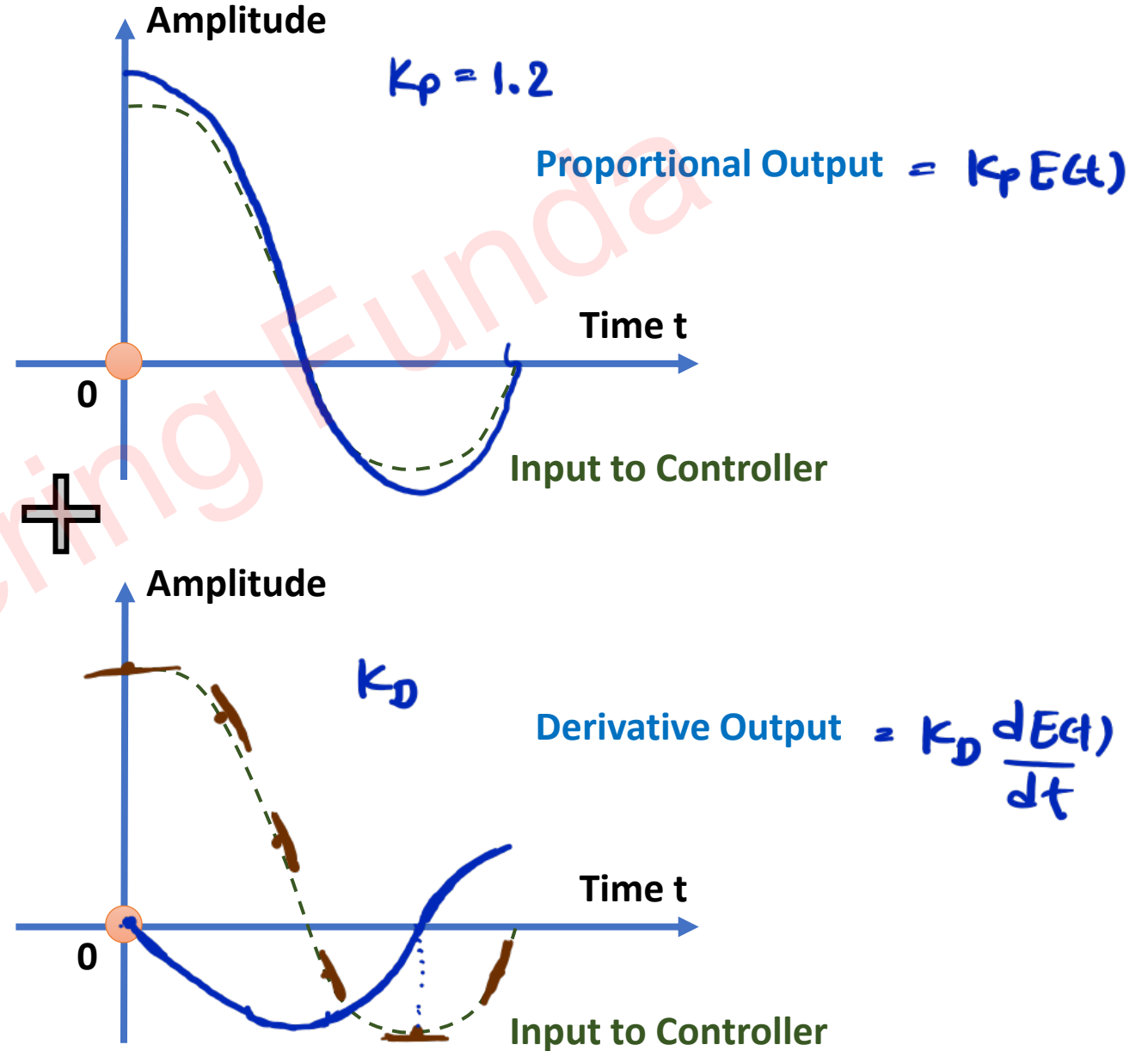
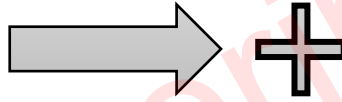
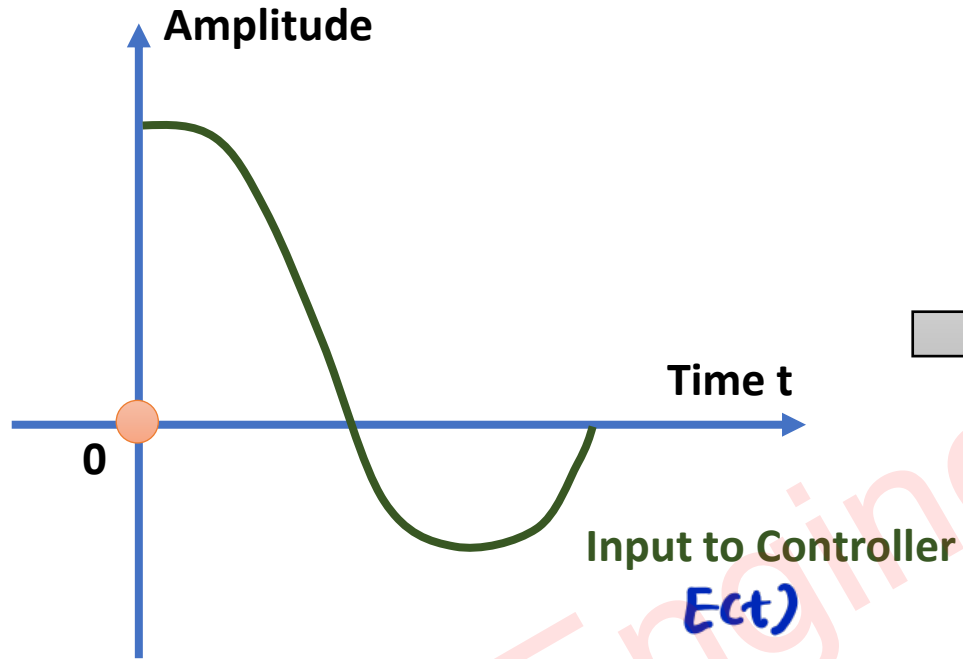
⇒ Apply Laplace X' form.

$$\Rightarrow M(s) = K_P E(s) + K_D (s E(s))$$

$$\Rightarrow M(s) = E(s) (K_P + s K_D)$$

$$\Rightarrow \frac{M(s)}{E(s)} = K_P + s K_D$$

Physical Understanding of PD Controller

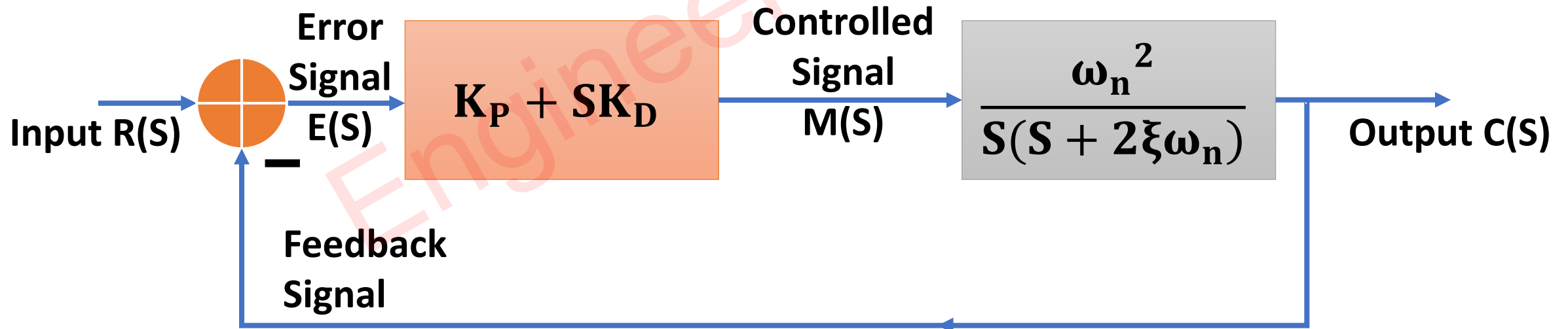


Significance of PD Controller

- ❑ Error Reduction (Due to Proportional Controller).
- ❑ Anticipatory Action (Due to Derivative Controller).
- ❑ Improvement in Dynamic Response (Lower Settling Time and Higher Stability Margin).
- ❑ Faster Transient Response.
- ❑ Reduced Overshoot and Oscillations.
- ❑ Improved Stability (Due to Derivative Controller).

$$\approx \frac{dE(t)}{dt} = SE(s)$$

Block Diagram of PD Controller with 2nd Order System



~ Transfer Function

$$\sim H(s) = 1$$

$$\sim G(s) = \frac{(K_p + sK_D)\omega_n^2}{s(s + 2\xi\omega_n)}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{\frac{(K_p + sK_D)\omega_n^2}{s(s + 2\xi\omega_n)}}{1 + \frac{(K_p + sK_D)\omega_n^2}{s(s + 2\xi\omega_n)}}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{(K_p + sK_D)\omega_n^2}{s(s + 2\xi\omega_n) + (K_p + sK_D)\omega_n^2}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{(K_p + sK_D)\omega_n^2}{s^2 + s(2\xi\omega_n + K_D\omega_n^2) + K_p\omega_n^2}$$

Advantages of PD Controller

- ☐ Improved Stability
- ☐ Faster Transient Response
- ☐ Reduced Overshoot and Oscillations
- ☐ Lower Settling Time

Disadvantages of PD Controller

- ☐ Steady State Error
- ☐ Sensitive to Noise
- ☐ Careful Tuning

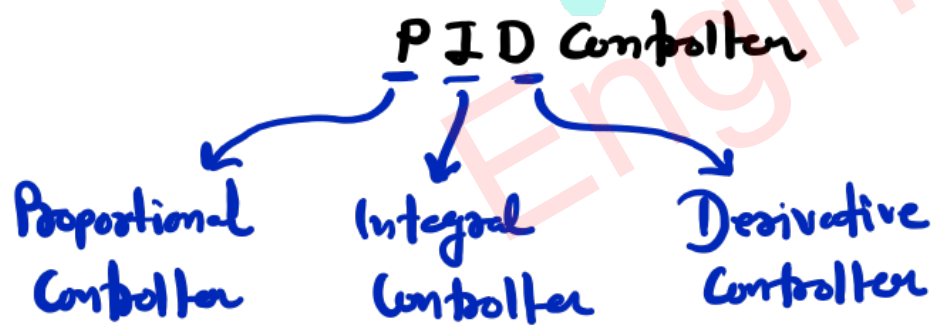
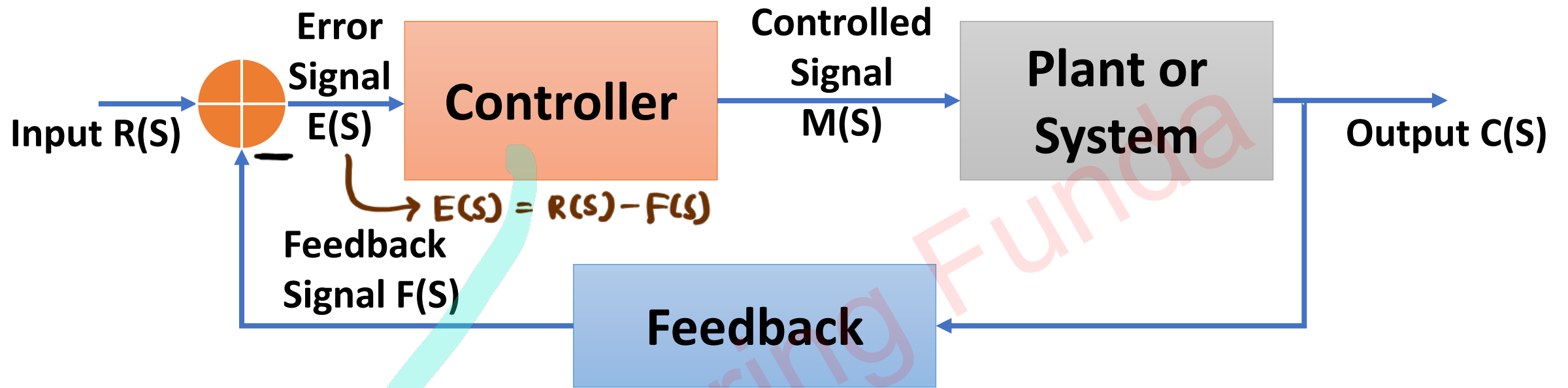
Engineering Funda

PID Controller

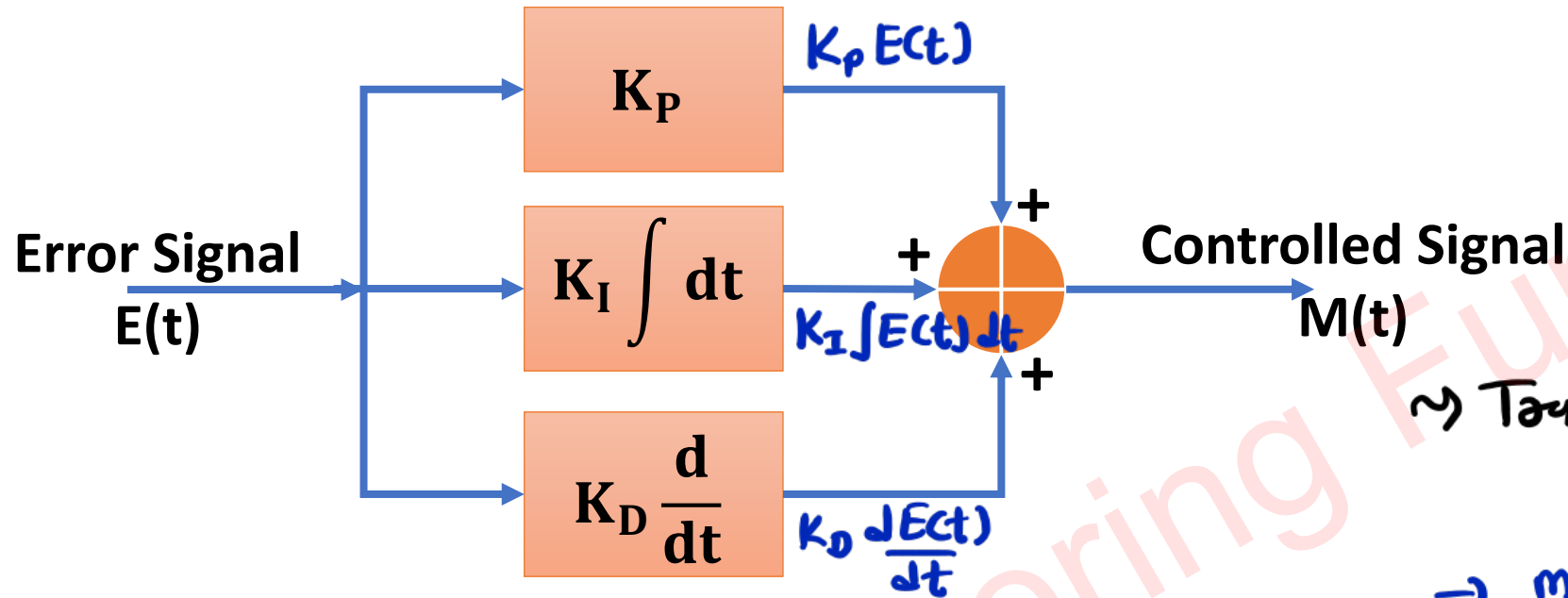
Outlines

- ☐ Basics of PID Controller
- ☐ Output of PID Controller
- ☐ Block Diagram of PID Controller
- ☐ Transfer Function of PID Controller
- ☐ Physical Understanding of PID Controller
- ☐ Significance of PID Controller
- ☐ Pros and Cons of PID Controller

Basics of PID Controller



Output of PID Controller



⇒ Transfer Function of PID Controller is given by:

$$\Rightarrow \frac{M(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s$$

⇒ Output of PID Controller is given by:

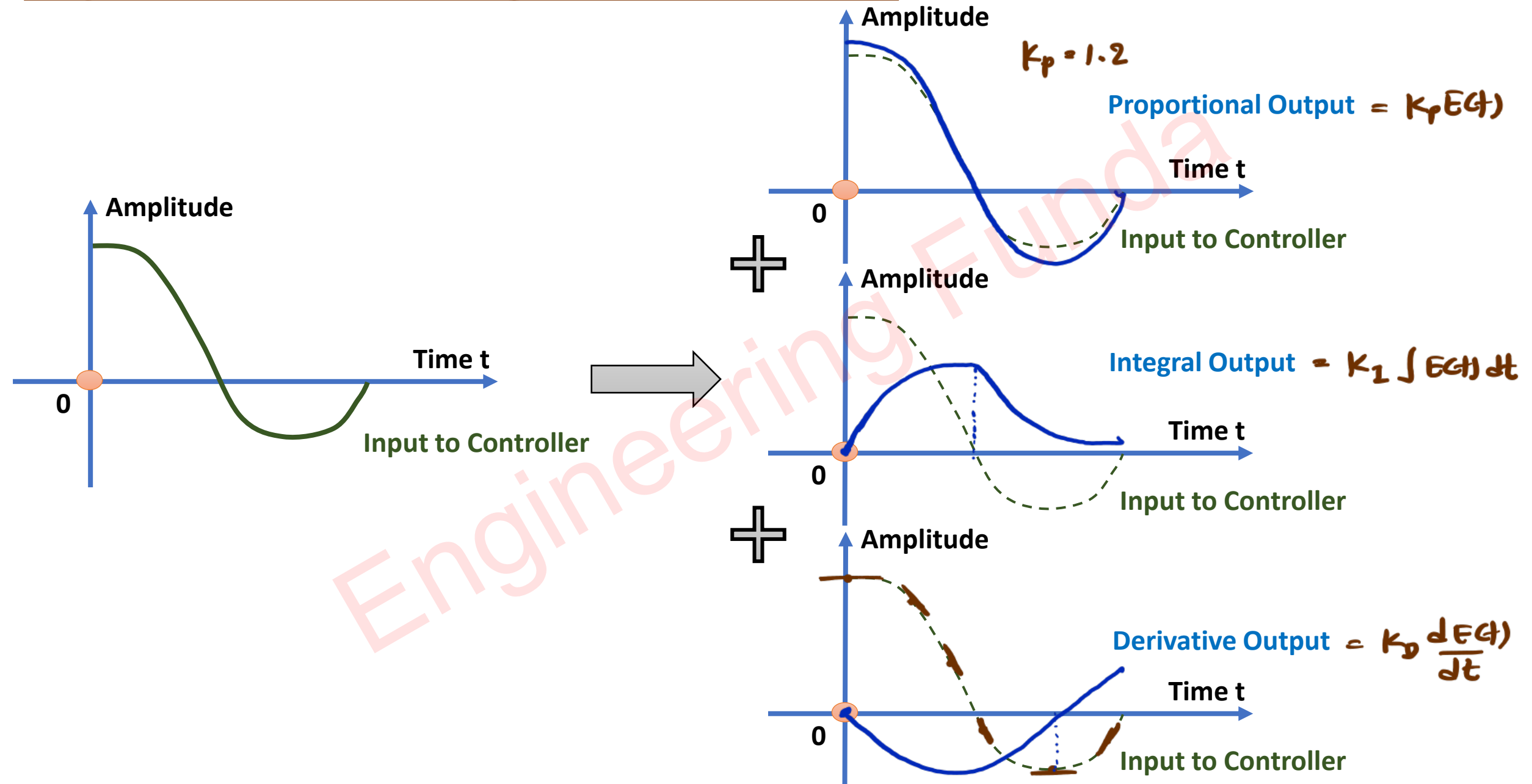
$$\Rightarrow m(t) = K_P E(t) + K_I \int E(t) dt + K_D \frac{dE(t)}{dt}$$

⇒ Apply Laplace Transform

$$\Rightarrow M(s) = K_P E(s) + K_I \frac{E(s)}{s} + K_D E(s) s$$

$$\Rightarrow M(s) = E(s) \left[K_P + \frac{K_I}{s} + s K_D \right]$$

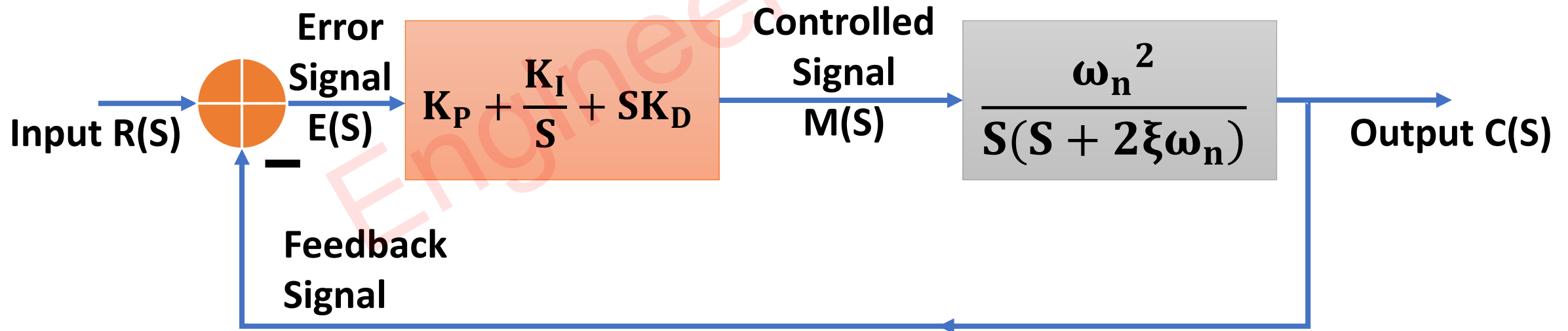
Physical Understanding of PID Controller



Significance of PID Controller

- ❑ No Steady State Error (Due to Integral Controller).
- ❑ Anticipatory Action (Due to Derivative Controller).
- ❑ Reduced Overshoot and Oscillations. (Derivative Controller)
- ❑ Lower Settling Time and Higher Stability Margin (Proportional and Derivative Controller).
- ❑ Faster Transient Response (Proportional and Derivative Controller).
- ❑ Improved Stability (Due to Derivative Controller).

Block Diagram of PID Controller with 2nd Order System



~> Transfer Function

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\sim H(s) = 1$$

$$\begin{aligned} \sim G(s) &= \left(K_p + \frac{K_1}{s} + K_D s \right) \left(\frac{\omega_n^2}{s(s + 2\xi\omega_n)} \right) \\ &= \frac{\omega_n^2 (K_D s^2 + K_p s + K_1)}{s^2 (s + 2\xi\omega_n)} \end{aligned}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{\frac{\omega_n^2 (K_D s^2 + K_p s + K_1)}{s^2 (s + 2\xi\omega_n)}}{1 + \frac{\omega_n^2 (K_D s^2 + K_p s + K_1)}{s^2 (s + 2\xi\omega_n)}}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{\omega_n^2 (K_D s^2 + K_p s + K_1)}{s^2 (s + 2\xi\omega_n) + \omega_n^2 (K_D s^2 + K_p s + K_1)}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{\omega_n^2 (K_D s^2 + K_p s + K_1)}{s^3 + s^2 (2\xi\omega_n + K_D \omega_n^2) + s (K_p \omega_n^2) + K_1 \omega_n^2}$$

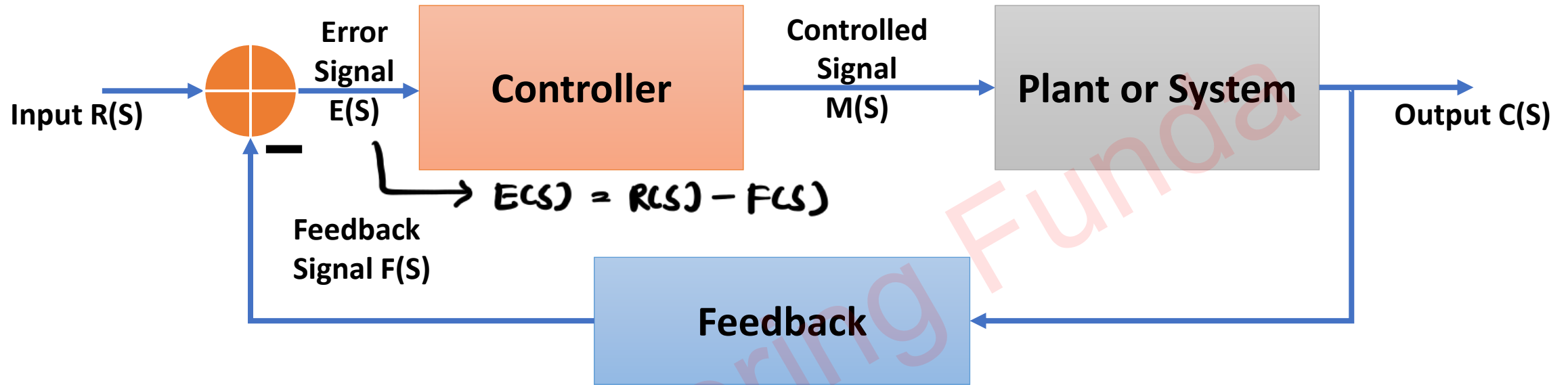
Advantages of PID Controller

- ❑ Proportional Controller – (Lower Rise Time, Fast Response at high K_p)
- ❑ Integral Controller – (Eliminate Steady State Error)
- ❑ Derivative Controller – (Improved Stability, Lower Overshoot and Lower Settling Time)
- ❑ Adoptability – (PID)

Disadvantages of PID Controller

- ❑ Tuning Complexity
- ❑ System Complexity

Comparison of P, PI, PD & PID Controller



Parameters	P Controller	PI Controller	PD Controller	PID Controller
❑ Controller Equation in Time Domain $M(t)$	$= K_p E(t)$	$= K_p E(t) + K_i \int E(t) dt$	$= K_p E(t) + K_d \frac{dE(t)}{dt}$	$= K_p E(t) + K_i \int E(t) dt + K_d \frac{dE(t)}{dt}$
❑ Controller Equation in S-Domain $M(S)$	$= K_p E(S)$	$= K_p E(S) + K_i \left(\frac{E(S)}{S} \right)$	$= K_p E(S) + K_d (E(S) S)$	$= K_p E(S) + K_i \left(\frac{E(S)}{S} \right) + K_d (E(S) S)$
❑ System Filtration	linear N/ω	LPF	HPF	= BPF or BRF

Parameters	P Controller	PI Controller	PD Controller	PID Controller
☐ Advantages	- K_p Gain Control Speed.	- Zero OFFSET (Steady State Error). - Improved Damping	- Decreased M_p - Decreased T_r - Decreased T_s - Large B.W.	- K_p (Control T_r) - K_i (X S.S.E.) - K_d (M_p & T_s)
☐ Drawbacks	- OFFSET or Steady State Error.	- Slow Response. - Stability	- OFFSET or Steady State Error. - Tuning Complexity	- Complex System - Tuning Complexity